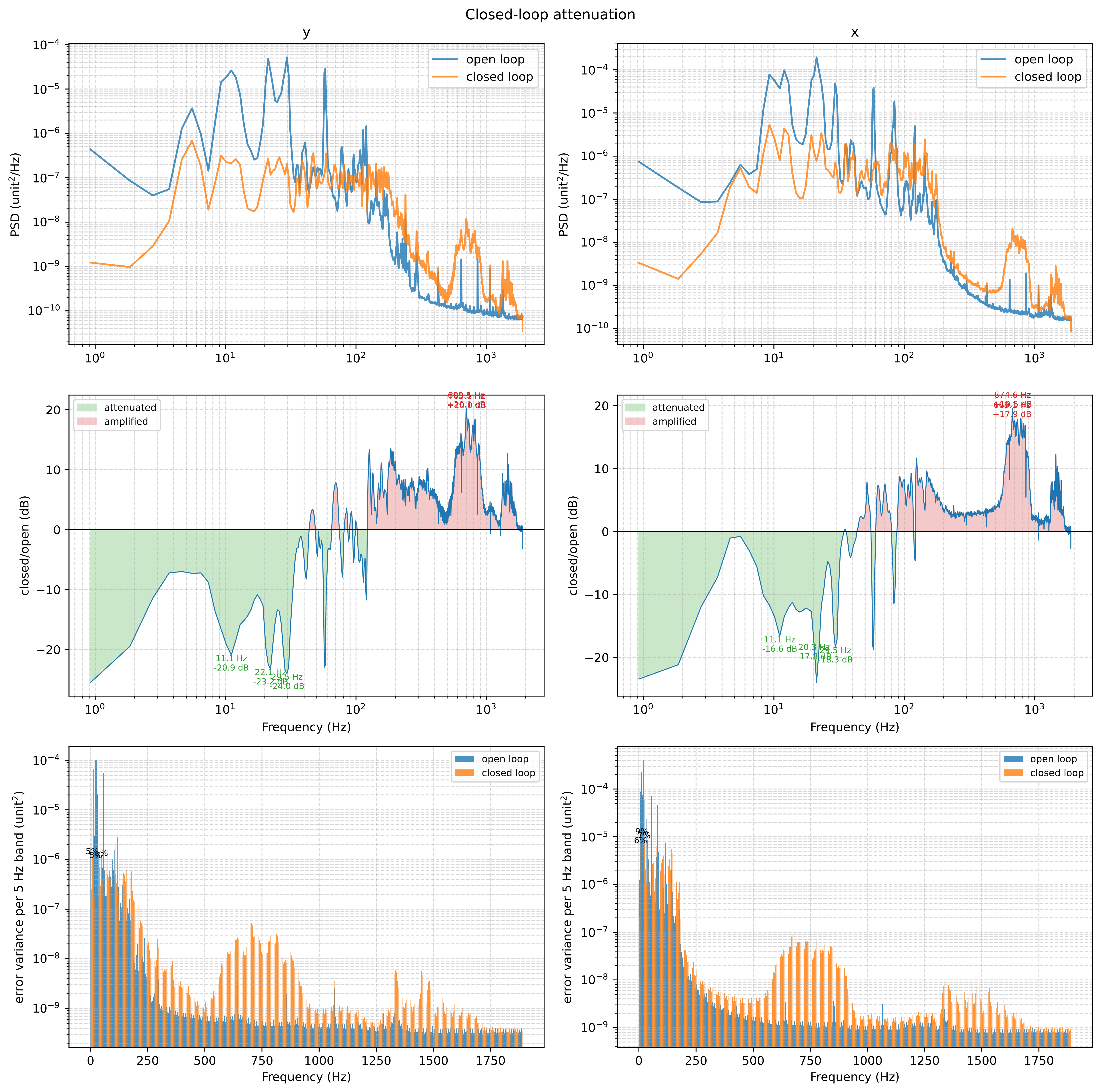

Closed-loop attenuation
y
x
open loop
closed loop
PSD (unit²/Hz)
attenuated
amplified
closed/open (dB)
Frequency (Hz)
11.1 Hz
-20.9 dB
11.1 Hz
-16.6 dB
674.6 Hz
+17.9 dB
error variance per 5 Hz band (unit²)

Run configuration and per-band RMS

```
Started: 2026-07-17 09:15:05
pixel_scale: 15.5 px per output unit
y: mean -0.062 -> -0.000 px;  RMS about mean 0.404 -> 0.073 px   (open -> closed)
x: mean +7.148 -> -0.000 px;  RMS about mean 0.693 -> 0.168 px   (open -> closed)

Config used:
ATTENUATION11 FSP RETEST -- new laser alignment, camera rate, and current Bode plant gains
Same continuously adapting full-band FSP as the successful attenuation11 run: broad_order=512, broad_mu=.03, start_delay=2 s, ramp=10 s; no frozen-weights test mode or safety-trip tuning
Camera/CoM match steering_tuned.json: 32x32 ROI, exposure 50, gain 100, 20x20 tracked window, threshold 22; coarse offsets 1.95/2.02 V
Current Bode plots fsm_bodex_716/fsm_bodey_716 give low-frequency normalized gains Kx=.868 and Ky=.630
Loop delay from the bode phase fits (5-300 Hz slope + 1 hidden frame): x 1.484 ms = 5.62 fr, y 1.681 ms = 6.37 fr at 3788 Hz. PER-AXIS delays: y global 6+0.0 (validated), x override 5+0.62 (bode fit; 5x latency runs 21:22-21:26 give departure 1.06 ms / 50% 1.32 ms = 5.00 fr every run). Departure-latency 4.02 fr was 1.6-2.3 fr short and caused the 310 Hz spiral of 07-16 20:55
All remaining observer and modal parameters match attenuation11; output is reported at 15.5 px per normalized unit
refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010
```

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.0710	0.0186	0.1439	0.0409
10-20	0.1297	0.0160	0.2698	0.0587
20-30	0.2193	0.0214	0.3333	0.0506
30-40	0.0731	0.0145	0.0872	0.0403
40-50	0.0208	0.0183	0.0328	0.0316
50-60	0.1149	0.0179	0.1317	0.0337
60-70	0.0139	0.0151	0.0157	0.0290
70-80	0.0204	0.0149	0.0317	0.0433
80-90	0.0150	0.0153	0.1110	0.0511
90-100	0.0175	0.0153	0.0135	0.0337
100-110	0.0228	0.0151	0.0212	0.0421
110-120	0.0342	0.0132	0.0427	0.0367
120-130	0.0091	0.0151	0.0184	0.0333
130-140	0.0076	0.0150	0.0146	0.0359
140-150	0.0101	0.0156	0.0258	0.0491
150-160	0.0058	0.0152	0.0095	0.0247
160-170	0.0067	0.0135	0.0089	0.0198
170-180	0.0073	0.0156	0.0109	0.0206
180-190	0.0024	0.0096	0.0048	0.0086
190-200	0.0018	0.0071	0.0034	0.0054
200-210	0.0027	0.0072	0.0028	0.0042
210-220	0.0019	0.0049	0.0023	0.0034
220-230	0.0022	0.0049	0.0023	0.0032
230-240	0.0029	0.0055	0.0022	0.0031
240-250	0.0015	0.0030	0.0018	0.0024
250-260	0.0014	0.0030	0.0019	0.0026
260-270	0.0009	0.0021	0.0016	0.0022
270-280	0.0009	0.0020	0.0015	0.0021
280-290	0.0011	0.0020	0.0015	0.0020
290-300	0.0017	0.0032	0.0015	0.0020
total 0-300	0.3036	0.0701	0.4982	0.1656